Bayesian Analysis of Animal Shelter Adoption Outcomes

A Bayesian analysis of Long Beach Animal Shelter intake and outcome records

Sadiqun Nur Ayan

Table of contents

1 Project Overview

Animal shelters collect intake and outcome records that can be used to understand which animals are more or less likely to be adopted. This project uses Long Beach Animal Shelter intake and outcome data to model the probability that a shelter record results in adoption.

The goal is not to make causal claims. Instead, the analysis estimates associations between adoption outcomes and observable animal or intake characteristics such as animal type, recorded sex/status, intake condition, intake type, and age at intake.

2 Research Question

Which animal and intake characteristics are associated with the probability of adoption among common adoptable pets at Long Beach Animal Shelter?

3 Data Source and Initial Structure

The analysis uses shelter intake and outcome records from Long Beach Animal Shelter. The dataset contains information on animal characteristics, intake details, and recorded shelter outcomes.

Table 1: Initial structure of the raw Long Beach Animal Shelter dataset

Metric	Value
Number of records	29,787
Number of columns	22
Unique animal IDs	28,754
Additional records from repeated animal IDs	1,033
Exact duplicate rows	0

The raw dataset contains shelter-level intake and outcome records. Because some `animal_id` values appear more than once, the rows are treated as shelter records rather than guaranteed unique animals throughout the analysis.

4 Data Cleaning and Outcome Definition

The response variable is defined as whether a shelter record resulted in adoption. Records with missing outcome type are removed because the adoption outcome is not observed. The analysis is restricted to common adoptable pet types so that the model focuses on a practical shelter adoption setting rather than combining very different animal categories.

Table 2: Summary of cleaned modelling dataset

Metric	Value
Raw records	29,787
Clean modelling records	23,638
Adopted records	6,106
Non-adoption records	17,532
Adoption rate	25.8%

After cleaning, the modelling dataset contains the records used for both exploratory analysis and Bayesian logistic regression. The binary response distinguishes adoption from all other observed shelter outcomes, so the results should be interpreted as associations with recorded adoption rather than with overall animal welfare success.

5 Exploratory Analysis

Before fitting the Bayesian logistic regression model, the cleaned data were explored to understand the adoption outcome distribution and how adoption rates vary across key animal and intake characteristics.

5.1 Adoption Outcome Distribution

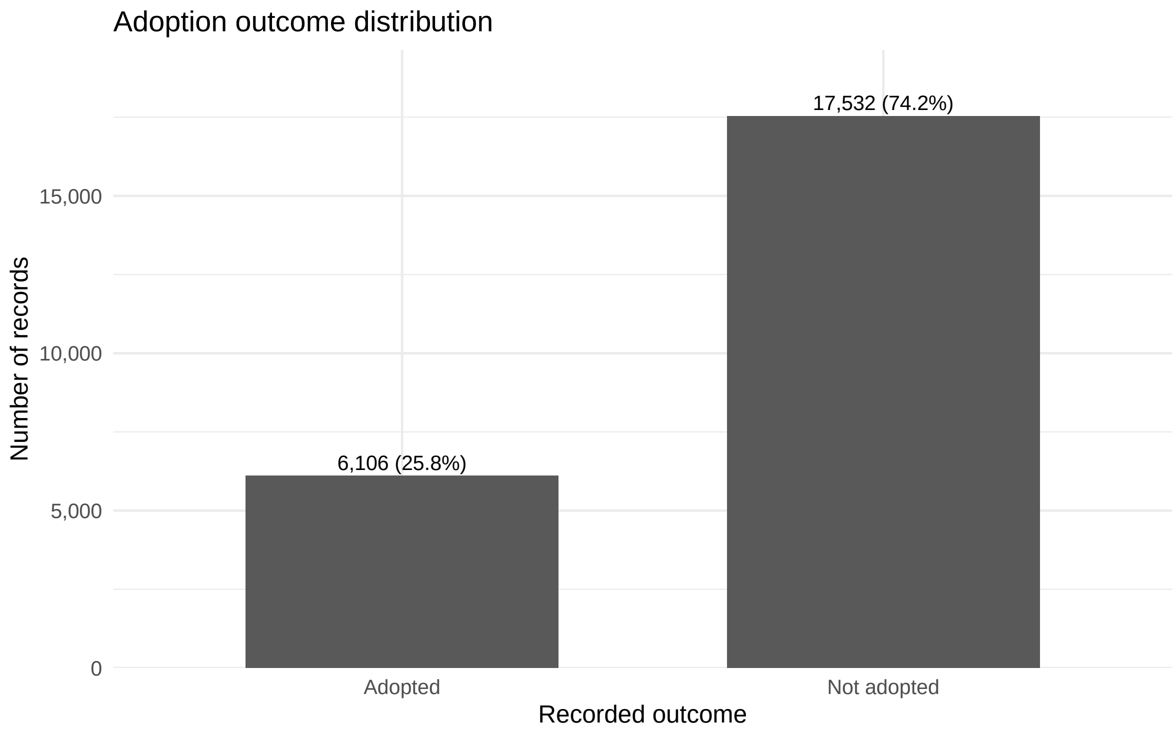

Figure 1: Distribution of adoption and non-adoption records in the cleaned modelling dataset.

The cleaned modelling dataset contains 6,106 adoption records and 17,532 non-adoption records, corresponding to an adoption rate of 25.8%. This confirms that adoption is the minority outcome in the cleaned data. As a result, the Bayesian model estimates adoption probability relative to a larger baseline group of other observed shelter outcomes.

5.2 Observed Adoption Rate by Animal Type

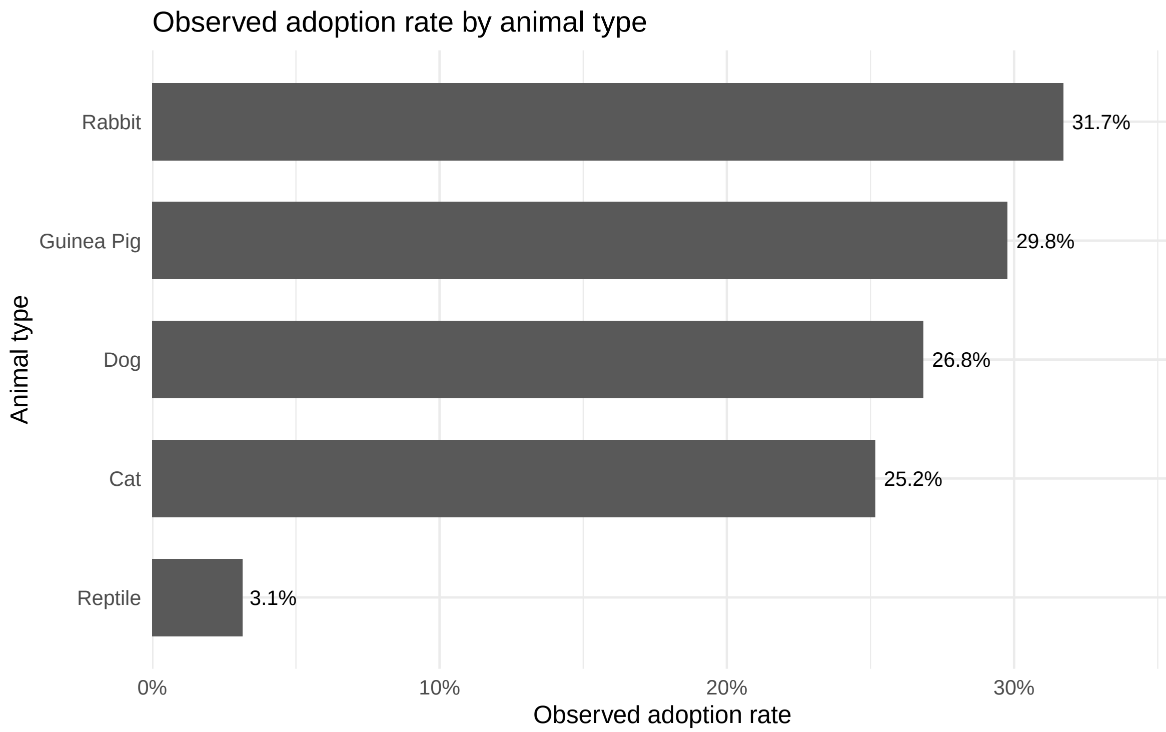

Figure 2: Observed adoption rate by animal type.

Observed adoption rates vary across animal types. Rabbits have the highest observed adoption rate at 31.7%, followed by guinea pigs at 29.8%, dogs at 26.8%, and cats at 25.2%. Reptiles have a much lower observed adoption rate at 3.1%. These are unadjusted rates, so they describe raw differences by animal type before controlling for other characteristics such as age, intake condition, intake type, and recorded sex/status.

5.3 Observed Adoption Rate by Age Group

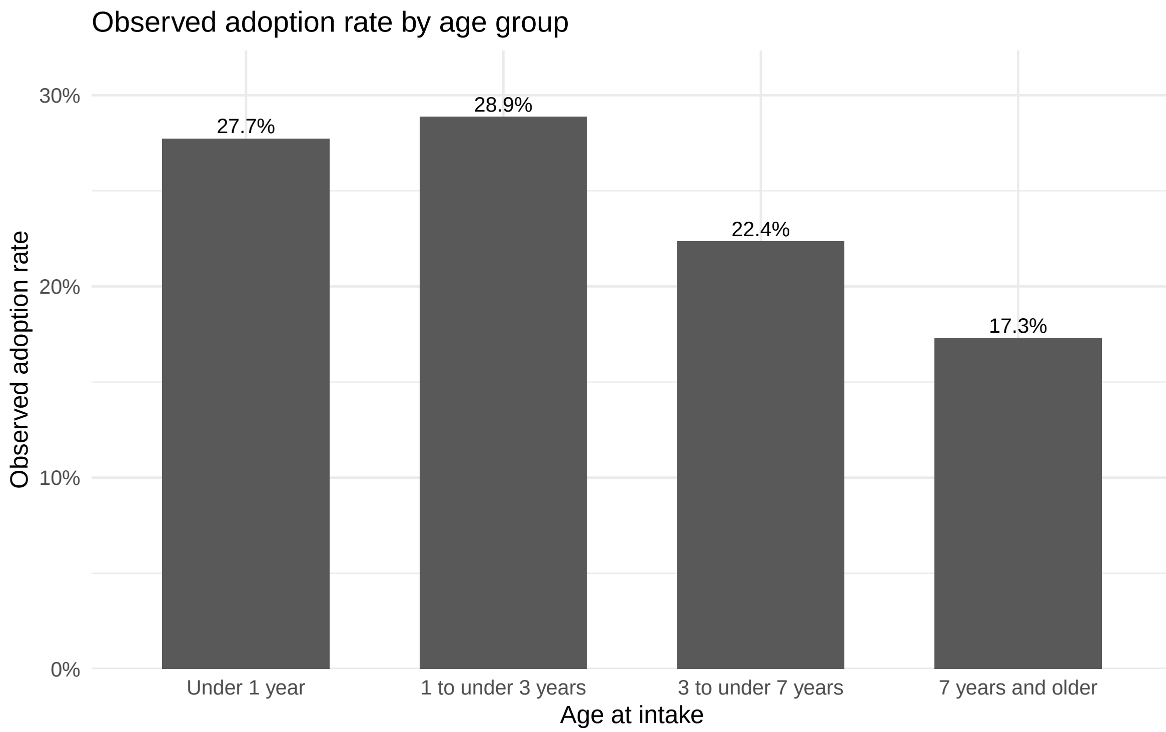

Figure 3: Observed adoption rate by age group.

Observed adoption rates decline for older animals. Records for animals under 3 years old have adoption rates around 28%, while records for animals aged 7 years and older have a lower observed adoption rate of 17.3%. This suggests age may be an important predictor in the Bayesian logistic regression model, although these are still unadjusted rates and do not control for animal type, intake condition, intake type, or recorded sex/status.

5.4 Observed Adoption Rate by Recorded Sex/Status

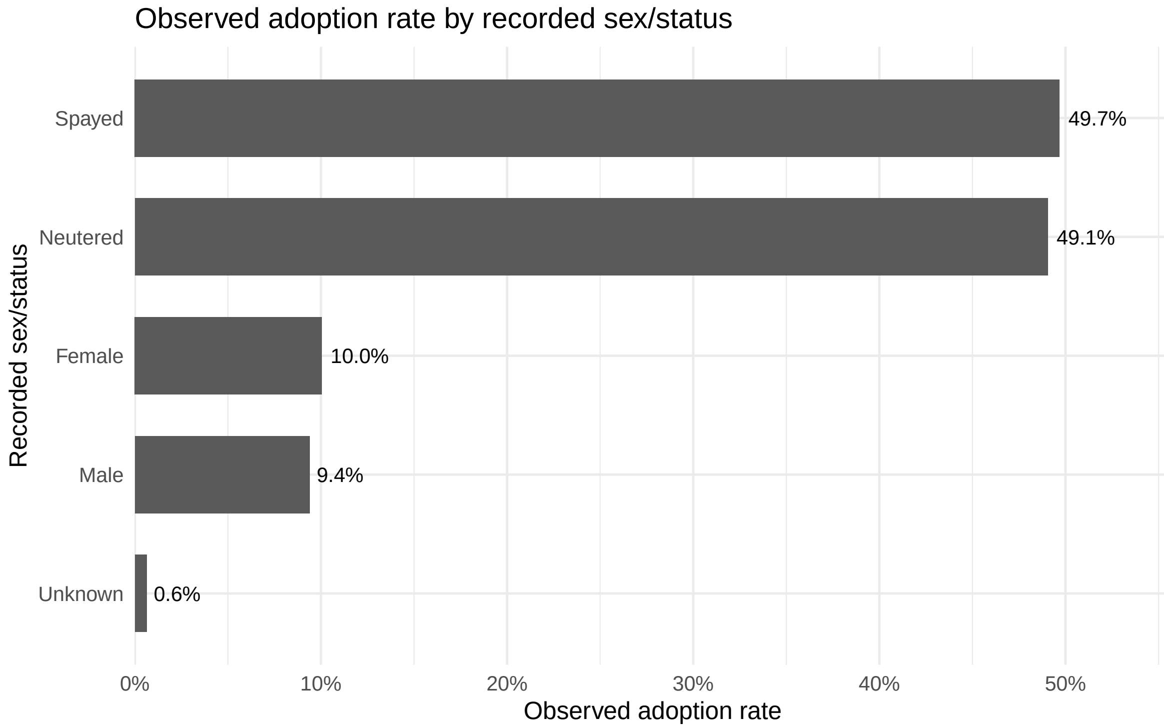

Figure 4: Observed adoption rate by recorded sex/status.

Observed adoption rates differ strongly by recorded sex/status. Records marked as spayed or neutered have adoption rates around 50%, while records marked only as female or male have adoption rates around 10%. Records with unknown sex/status have the lowest observed adoption rate at 0.6%. This variable should be interpreted carefully because it combines biological sex with sterilisation status and may partly reflect shelter processing before adoption.

6 Bayesian Model Specification

The response variable is binary, where $Y_i = 1$ indicates that shelter record i resulted in adoption and $Y_i = 0$ indicates another observed shelter outcome. A Bayesian logistic regression model is used because the outcome is binary and the goal is to estimate how animal and intake characteristics are associated with adoption probability.

For each shelter record i:

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\text{logit}(p_i) = \mathbf{x}_i^\top \beta$$

where p_i is the probability of adoption and $\mathbf{x}_i$ contains the predictors for animal type, recorded sex/status, intake condition, intake type, and scaled age at intake.

Weakly informative normal priors are assigned to the regression coefficients:

$$\beta_j \sim \text{Normal}(0, 10^2)$$

These priors are centred at zero, representing no strong prior assumption about the direction of each association. The prior standard deviation of 10 is broad on the log-odds scale, allowing the data to dominate while still keeping the Bayesian model formally specified.

7 Modelling Data Preparation

Categorical predictors are converted to factors before constructing the logistic regression design matrix. Reference categories are selected to make coefficient interpretation clear: cats, female records, normal intake condition, and stray intake type are used as baseline groups. Age at intake is standardised so that its coefficient represents the effect of a one-standard-deviation increase in age.

Table 3: Summary of modelling sample and design matrix

Metric	Value
MCMC modelling records	7,090
Adopted records in MCMC sample	1,831
Non-adoption records in MCMC sample	5,259
Number of model predictors including intercept	25

The Bayesian model is fitted on a reproducible stratified sample of the cleaned modelling dataset. Stratified sampling preserves the adoption/non-adoption balance while reducing JAGS runtime, making the workflow easier to reproduce during report generation.

8 Bayesian Model Fitting with JAGS

The Bayesian logistic regression model was fitted using JAGS. The model uses a Bernoulli likelihood with a logit link and weakly informative normal priors for all regression coefficients.

```
model {
  for (i in 1:N) {
    Y[i] ~ dbern(p[i])
    logit(p[i]) <- inprod(X[i, ], beta[])
  }

  for (j in 1:P) {
    beta[j] ~ dnorm(0, 0.01)
  }
}
```

9 Posterior Summary and MCMC Diagnostics

Posterior convergence was assessed using Gelman-Rubin $\hat{R}$ values, effective sample sizes, and visual inspection of selected MCMC trace plots.

Table 4: MCMC convergence diagnostics for the Bayesian logistic regression model

Metric	Value
Number of MCMC chains	2
Total retained posterior draws	8,000
Maximum Rhat	1.001
Minimum effective sample size	670

The convergence diagnostics suggest that the MCMC sampling was acceptable for this analysis. The maximum $\hat{R}$ value was 1.001, which is very close to 1 and indicates good agreement between chains. The minimum effective sample size was 670, suggesting that the retained posterior draws provide enough independent information for stable posterior summaries. Trace plots are examined next as a visual diagnostic.

9.1 MCMC Trace Plots

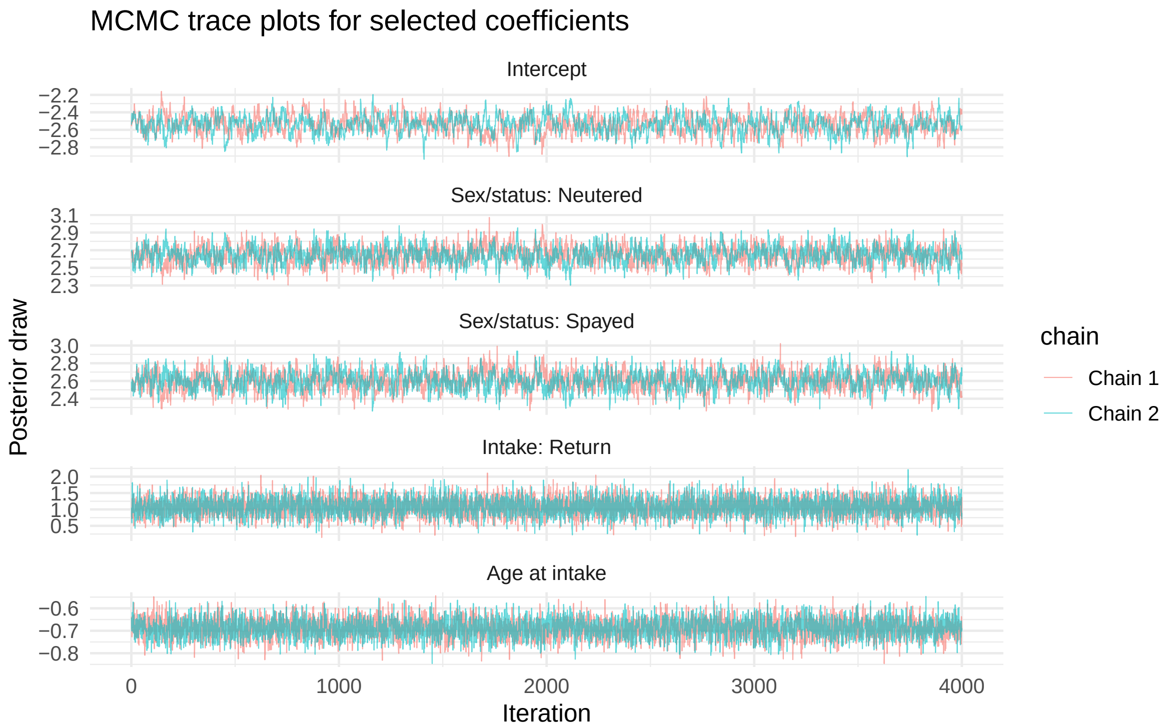

Figure 5: Trace plots for selected posterior coefficients.

The trace plots show stable sampling behaviour for the selected coefficients, with no clear long-term trends or systematic drift across iterations. Combined with the low maximum $\hat{R}$ value and acceptable effective sample size, these plots support the use of the posterior samples for coefficient interpretation and prediction.

9.2 Posterior Coefficient Effects

The posterior coefficient estimates are shown on the log-odds scale. Positive values indicate higher adoption log-odds relative to the reference category, while negative values indicate lower adoption log-odds. The reference categories are cat, female, normal intake condition, and stray intake type.

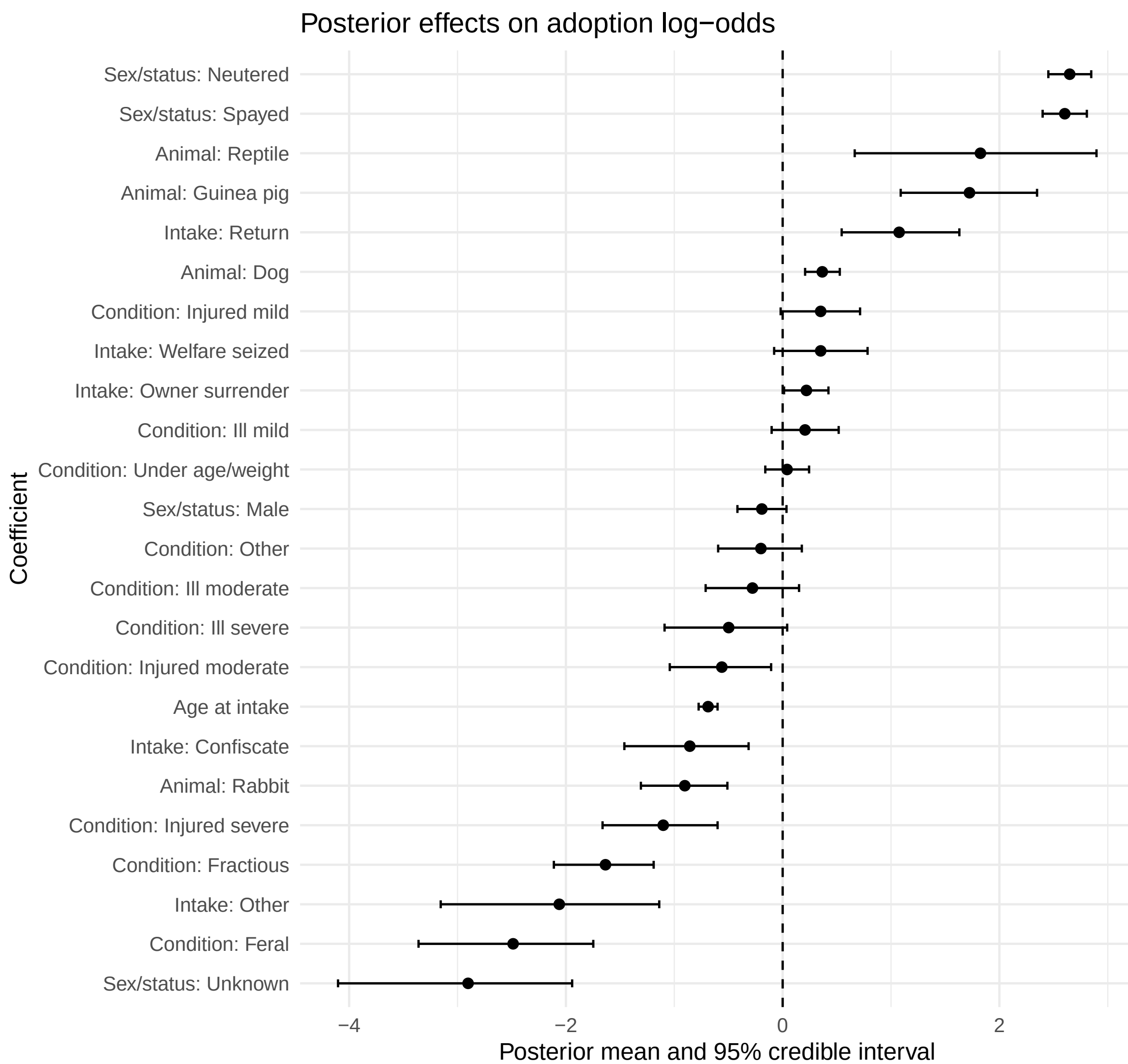

Figure 6: Posterior means and 95% credible intervals for Bayesian logistic regression coefficients.

The posterior effects show that recorded spayed and neutered status are strongly associated with higher adoption log-odds relative to the female reference category. Guinea pigs, reptiles, dogs, and return intake records also have positive posterior effects, although some of these categories have wider credible intervals. Older age at intake is associated with lower adoption log-odds. Records with unknown sex/status, feral or fractious intake condition, and other intake type show clearly negative posterior effects. Coefficients whose 95% credible intervals cross zero should be interpreted more cautiously because the posterior evidence for a directional association is weaker.

10 Predicted Adoption Probabilities

Posterior coefficients are on the log-odds scale, so selected model-based predictions are used to express results as adoption probabilities. The profiles below compare animal types and recorded sex/status while holding intake condition as normal, intake type as stray, and age at the median value.

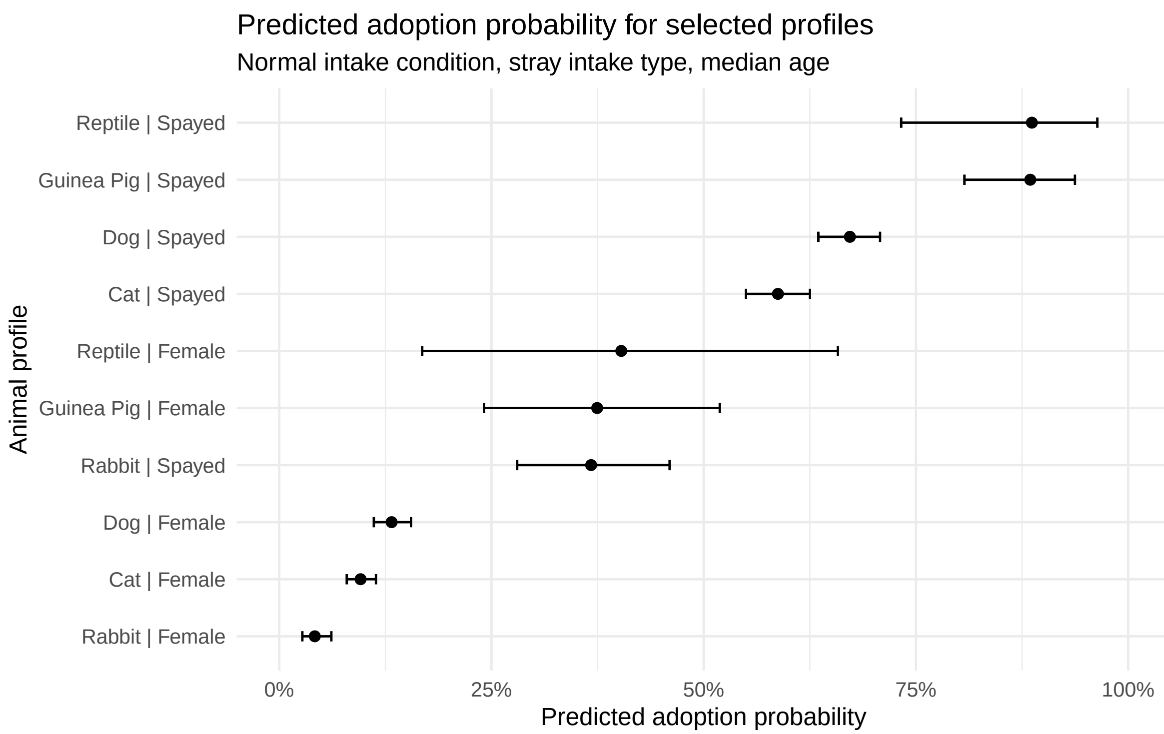

Figure 7: Posterior predicted adoption probabilities for selected animal profiles.

The predicted probabilities show that recorded spayed status is associated with substantially higher model-estimated adoption probability across animal types. Cats and dogs have relatively low predicted adoption probabilities when recorded as female only, but much higher probabilities when recorded as spayed. Guinea pigs and reptiles also show high predicted probabilities when recorded as spayed, although their intervals are wider, indicating greater posterior uncertainty. These predictions are conditional on normal intake condition, stray intake type, and median age.

11 Conclusion

This project used Bayesian logistic regression to estimate how animal and intake characteristics are associated with adoption outcomes in Long Beach Animal Shelter records. The model suggests that recorded sex/status, animal type, intake type, intake condition, and age at intake are all relevant to adoption probability.

The strongest positive associations were observed for records marked as spayed or neutered, while older age, unknown sex/status, feral or fractious intake condition, and other intake type were associated with lower adoption log-odds. Model-based predicted probabilities also showed that the same animal type can have substantially different adoption probabilities depending on recorded sex/status.

These results should be interpreted as associations rather than causal effects. In practice, the analysis can help identify groups of shelter records with lower model-estimated adoption probability, which may support more targeted outreach, resource planning, and adoption support strategies.